# * Practical-1 *

* Write a program to implement Single Player Game (using Heuristic Function)

* tic-tac-toe.py

```
theBoard = {
    '7' : ' ' , '8' : ' ' , '9' : ' ' ,
    '4' : ' ' , '5' : ' ' , '6' : ' ' ,
    '1' : ' ' , '2' : ' ' , '3' : ' ' ,
}

boardKeys = list(theBoard.Keys())

def printBoard (board):
    print (board ['7'] + '|' + board['8'] +
                          '|' + board['9'])
    print ('-+-+-')
    print (board ['4'] + '|' + board ['5'] +
                          '|' + board ['6'])
    print ('-+-+-')
    print (board ['1'] + '|' + board ['2'] +
                          '|' + board ['3'])

def game ():
    turn = 'X'
    count = 0
```

```
for i in range(10):
    printBoard(theBoard)
    print("It's your turn," + turn
          + ". Move to which place?")

    move = input()

    if theBoard[move] == ' ':
        theBoard[move] = turn
        count += 1
    else:
        print("That place is alread
              filled.\n Move to
              which place?")
        continue

    if count >= 5:
        if theBoard['7'] == theBoard['8']
            == theBoard['9'] != ' ':
        printBoard(theBoard)
        print("\nGame Over.\n")
        print(" *** " + turn + " won")
        break

    elif theBoard['4'] == theBoard['5']
            == theBoard['6'] != ' ':
        printBoard(theBoard)
        print("\nGame Over.\n")
        print("***" + turn + " won ")
        break
```

```
elif theBoard['1'] == theBoard['2']
    == theBoard['3'] != ' ':
    printBoard(theBoard)
    print("\nGame Over.\n")
    print("*** "+ turn + " won ")
    break

elif theBoard['1'] == theBoard['4']
    == theBoard['7'] != ' ':
    printBoard(theBoard)
    print("\nGame Over.\n")
    print("*** "+ turn + " won")
    break

elif theBoard['2'] == theBoard['5']
    == theBoard['8'] != ' ':
    printBoard(theBoard)
    print(" \nGame Over.\n")
    print(" ***" + turn + " won")
    break

elif theBoard['3'] == theBoard['6']
    == theBoard['9'] != ' ':
    printBoard(theBoard)
    print("\nGame Over.\n")
    print("*** " + turn + " won")
    break
```

```
            elif theBoard['7'] == theBoard['5']
                == theBoard['3'] != ' ':
                printBoard(theBoard)
                print("\nGame Over.\n")
                print("***" + turn + "won")
                break

            elif theBoard['1'] == theBoard['5']
                == theBoard['9'] != ' ':
                printBoard(theBoard)
                print("\nGame Over.\n")
                print("***" + turn + "won")
                break.

        if count == 9:
            print("\nGame Over.\n")
            print("It's a Tie!!")

        if turn == 'X':
            turn = 'O'
        else:
            turn = 'X'

restart = input("Do want to play Again? (y/n):")
    if restart.lower() == 'y':
        for key in board_keys:
            theBoard[key] = " "
        game()

__name__ == "__main__":
game()
```

# * Output *

```
 | |
-+-+-
 | |
-+-+-
 | |
```

It's your turn, X.
Move to which place?

5

```
 | |
-+-+-
 |X|
-+-+-
 | |
```

It's your turn, O.
Move to which place?

1

```
 | |
-+-+-
 |X|
-+-+-
O| |
```

9

```
 | |X
-+-+-
 |X|
-+-+-
O| |
```

2

```
 | |X
-+-+-
 |X|
-+-+-
O|O|
```

1

```
 | |X
-+-+-
 |X|
-+-+-
O|O|
```

3

Game Over.

*** X won. ***

Do you want to play Again?(y/n):n